

Right-to-Repair & Circular Economy Platform

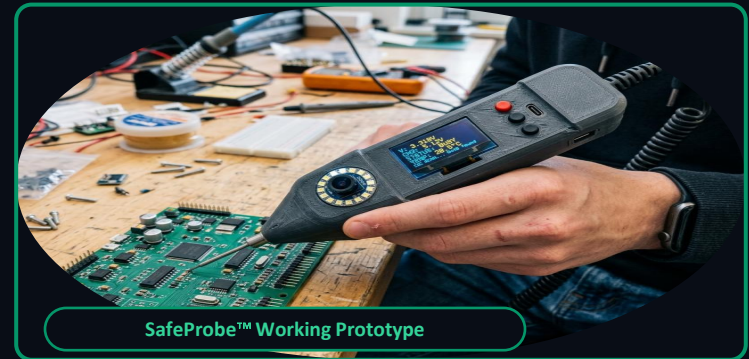
FixGrid & SafeProbe™

The Complete Ecosystem: A Live Web Platform (www.vytron.me) & A Frugal Handheld Diagnostic Wand Championing Sustainable Electronics Repair

✓ Live Digital Platform: Neighborhood search, escrow & warranties

✓ Physical Innovation: Under ₹1,200 SafeProbe™ diagnostic wand

✓ Dual-Loop QR Warranty: Tamper-evident seals with photo proof



The Two-Pillar Repair Ecosystem

PILLAR 1: WEB PLATFORM

FixGrid Digital Trust Web Network

- Live Web Platform: Hyper-local directory with real-time shop hours and 21 repair categories.
- Smart Escrow: Funds held securely and released only after verified customer satisfaction.
- Platform-Backed Warranty: 5-day standard warranty + digital shopkeeper warranty records.
- Artisan Cashback: 5% completed-bill rebate rewarding honest local craftsmanship.

www.vytron.me

Live Web Platform

PILLAR 2: HARDWARE

SafeProbe™ Smart Diagnostic Wand

- Frugal Tool BOM: Built for under ₹1,200 (\$14), replacing ₹1.5L lab oscilloscopes.
- Instant Micro-Touch Sensing: Precision needle probe tests trace impedance with audio chirp & OLED color.
- 5MP Macro Camera Microscope: Snaps magnified photo proof sent straight to customer's phone.
- Dual-Loop QR Passport: Issues physical tamper-evident QR void seals linked to cloud warranty.

₹1,200 BOM

SafeProbe Hardware

5MP Macro

Customer Photo Proof

Zero E-Waste

Circular Economy Goal

The Throwaway Culture & The Diagnostic Void

1. The Trust & Visibility Void

- Consumers discard repairable electronics assuming replacement is the only safe option.
- Roadside repair artisans lack digital storefronts, standardized pricing, and formal warranty tracking.
- Consumers fear arbitrary overcharging, counterfeit parts, and shoddy workmanship.

2. The Hardware Diagnostic Void (Root Cause)

- Lab oscilloscopes & inspection microscopes cost ₹1,00,000 to ₹1,50,000—unaffordable for local mechanics.
- Crude multimeters cause accidental voltage spikes, frying delicate microprocessors.
- Micro-faults are invisible to naked eyes; without photo proof, consumers distrust technicians.

₹1.2L Cr

THE LOST VALUE

Billions wasted buying new replacements when only a ₹50 micro-component had failed.

85%+

SKILLED HEROES OVERLOOKED

Vast majority of neighborhood repairers remain informal, uncertified, and underpaid.

Millions

E-WASTE CATASTROPHE

Repairable electronics flood municipal landfills, generating toxic heavy metal contamination.

SafeProbe™: Lab Diagnostics in a Handheld Wand



SafeProbe™ Working Prototype (₹1,200 / \$14 BOM)

MICRO-TOUCH SENSING

Instant Fault Chirp

Technician touches probe needle to circuit trace. Conducts rapid impedance testing in milliseconds, beeping and lighting Green (Normal) or Red (Fault) on OLED screen. Safe sub-3.3V (<5mA) sensing prevents static shock.

MAGNIFIED VISUALS

5MP Macro Photo Proof

Integrated 5MP macro camera with polarized ring LEDs captures 10x close-up photos of burned ICs or cracked solder joints, sending undeniable visual proof straight to the customer's smartphone before soldering.

DIGITAL PASSPORT

Dual-Loop QR Warranty

Tool writes a physical tamper-evident QR void seal placed over gadget seam. Customer scans to review diagnostic readings, before/after photos, and activates a guaranteed 30-day digital warranty.

FixGrid Web Platform (www.vytron.me)

Hyper-Local Search & Map Engine

Allows consumers to discover verified repair shops within walking distance across 21 repair categories (Phones, Laptops, Appliances, Watches) with live opening-hour status.

- Next.js 16 + Leaflet Map Engine

Smart Escrow & Payment Protection

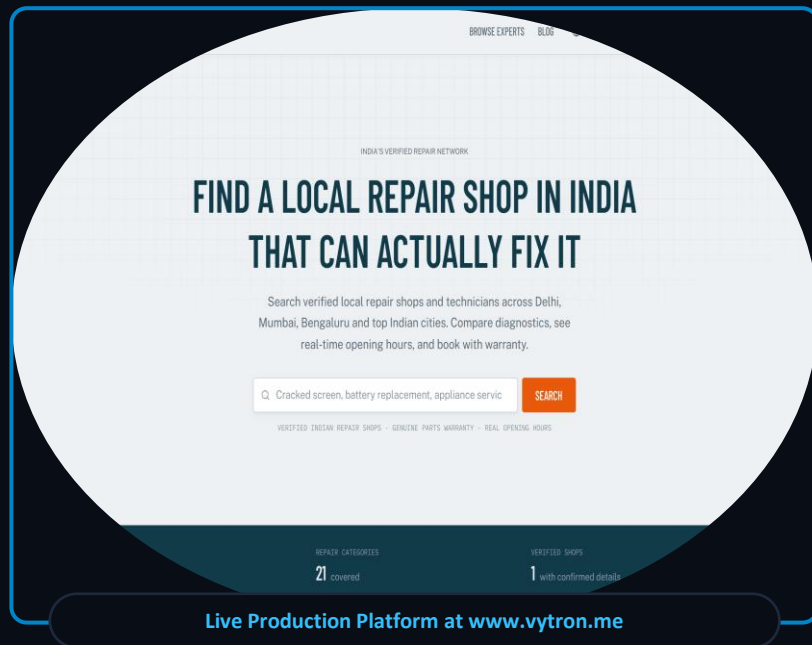
Customer payments are locked securely on the platform upon booking and released to the shopkeeper only after the customer inspects and signs off on verified satisfaction.

- Supabase PostgreSQL Triggers & RLS

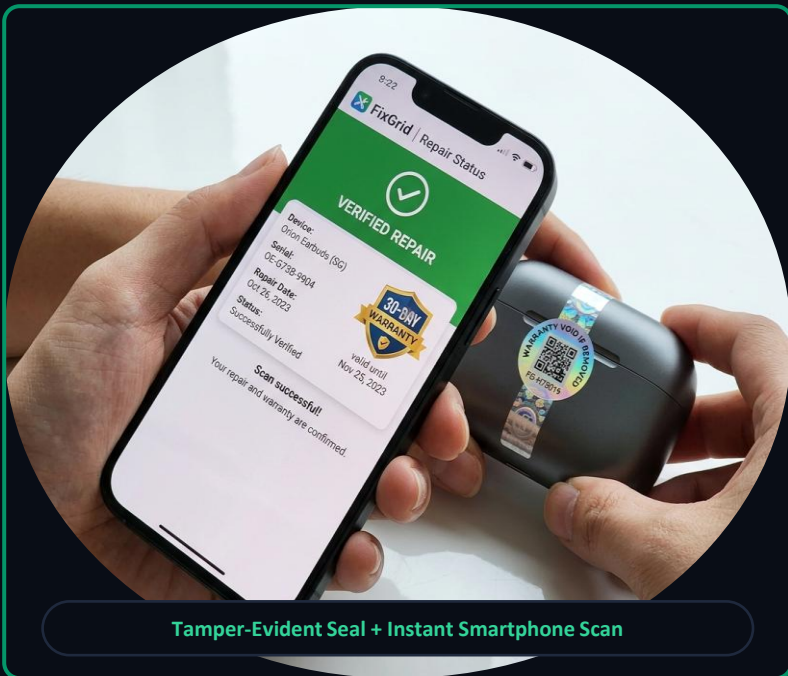
Artisan Storefronts & 5% Cashback

Provides informal repair mechanics with instant SEO-indexed digital profiles, verified reviews, diagnostic log uploads, and a 5% completed-bill cashback incentive.

- Shopkeeper Empowerment Engine



Dual-Loop QR Digital Product Passport



Tamper-Evident Seal + Instant Smartphone Scan

STEP 01

SafeProbe Diagnostics Upload

When the technician repairs the gadget, the SafeProbe wand syncs the passing impedance test and high-res macro photos directly to the FixGrid cloud database via ESP32 Wi-Fi/BLE.

STEP 02

Physical Holographic Void Seal

A serialized, tamper-evident holographic QR void sticker is printed and affixed over the device seam. If anyone attempts to tamper with or open the device, the physical pattern breaks.

STEP 03

Customer Scan & Warranty Activation

The customer scans the QR code with any standard smartphone camera. It displays component before/after photos, diagnostic health logs, and activates an ironclad 30-day digital warranty certificate.

Tapping the \$15.2B Repair Economy

60–80%**ECONOMIC (Consumer & Artisan)**

Consumers save 60–80% compared to purchasing new devices. Local mechanics gain formal digital identity, repeat customers, and higher daily earnings.

100%**SOCIAL (Community Dignity)**

Brings unorganized street artisans into the formal digital economy, elevating informal mechanics with lab-grade diagnostic tools and transparent trust.

Zero**ECOLOGICAL (Planet Circularity)**

Diverts hundreds of kilograms of toxic printed circuit boards and lithium batteries from municipal landfills, cutting manufacturing carbon emissions.

1. Platform Take-Rate (5–8%)

Competitive transaction fee on every escrow-verified repair booking.

2. Customer Care+ Subscription (₹199/mo | ₹1,499/yr)

Household repair pass: free SafeProbe diagnostics, zero booking fees & 60-day warranty.

3. Shop Pro SaaS & Hardware Tier (₹999/mo)

SafeProbe firmware updates, warranty management, digital store CRM & analytics.

4. Certified Spare Parts Logistics

Curated B2B marketplace revenue for verified authentic micro-components.

A Movement Towards a Sustainable Repair Economy

01

PHASE 1 (MONTHS 1–6)

Pilot Repair Clusters

- Deploy 50 SafeProbe units across Mumbai electronics repair markets.
- Validate web escrow & before/after macro photo flow with 1,000+ customer repairs.

02

PHASE 2 (MONTHS 7–12)

Community Drives

- Partner with schools & colleges for zero-e-waste community repair clinics.
- Roll out Customer Care+ subscription pass and certified technician onboarding.

03

PHASE 3 (YEAR 2+)

Municipal Scaling

- Integrate with municipal e-waste boards for certified gadget triage.
- Scale frugal SafeProbe tool manufacturing to local repairers nationwide.

FixGrid & SafeProbe™ Ecosystem

A Unified Web & Hardware Solution for Right-to-Repair, Consumer Trust & Circular Sustainability

Project FixGrid

www.vytron.me